IMPLEMENTATION FOLLOW-UP

DLSS Super Resolution fixes

The corrected integration audit supplied nine action items. This patch repairs the supported vanilla DLSS/DLAA path, adds missing particle motion and diagnostics, and tests the native boundary in game. It does not certify every procedural renderer, shader pack or possible temporal artifact.

Test system: Minecraft 26.3 / Fabric 0.19.5, RTX 3060 Ti, NVIDIA 591.86, user-imported Streamline 2.14.1 / DLSS runtime 310.9.1. Isolated test instance; normal runClient configuration and options preserved.

Audit item	Implemented result
F1 - particle motion	Match actual billboard translation, rotation and scale. Alpha-tested atlas coverage, scene-depth occlusion and nearest dominant translucent sample; adjustable opacity threshold.
F2 - unsupported mask	Remove modern DLSS bias-current-color tags, history/mask work, debug view and misleading controls. FSR retains its own masks.
F3 - geometry gaps / cap	Add block-model and custom quad/triangle capture, reject incomplete topology, expose coverage counters, offer 262K/1M/2M budgets and test overflow.
F4 - Vulkan resource metadata	Chain explicit mip/layer/aspect ranges on all four native resources; depth uses DEPTH, not the SDK default COLOR aspect.
F5 - color contract	Explicit source encoding; RGBA16F linear/HDR input/output and matching native flag, safe feature recreation on encoding changes, SDR/HDR probes.
F6 - texture validation	Compare DLAA/Quality on foliage, grids, signs, ores and a 64px checker pack. Keep terrain MIP bias optional rather than applying it blindly to unmipped textures.
F7 - camera metadata	Initialize pinhole offset to (0,0); keep the validated jitter/vector conventions. Guard invalid camera matrices before native dispatch.
F8 - honest telemetry	Label model as requested; separate SDK feature VRAM estimate from owned AA targets and expose motion coverage details.
F9 - lifecycle	Reset camera identity/perspective, render gaps and resource reloads. Retire cached pass buffers before device teardown; test resizing, dimensions, recovery and clean exit.

Additional validation fixes: normalize promoted Vulkan 1.2 feature structures, remove conflicting buffer-address extensions, enable supported unformatted storage-image writes and shaderInt16. Java and native bridge now use ABI 4.

REAL INPUT-BUFFER EVIDENCE

Particles now carry motion



Left: reconstructed flame particles. Right: corresponding input velocity magnitude, 0 to 0.8 input pixels. Both use screenshot orientation; raw GPU rows are flipped only for display. Nearest-neighbor enlargement preserves pixel evidence.

Capture / ROI	Nonzero moving input pixels	Captured matched billboards
Flame tracking disabled, 12 frames	0 in every frame	0
Flame tracking enabled, 24 frames	6,966 to 8,258	285 to 324
Flame with improved transparency	6,230 to 7,462	283 to 315
Smoke, dominant opacity policy	3,032 to 4,149	226 to 262
Smoke behind opaque wall	0 in every ROI frame	241 to 277

ROI: normalized screen x=0.31..0.70, y=0.38..0.62. Nonzero means speed above 0.001 input pixels. The occluded case still captures particles, but depth testing prevents them from overwriting visible world motion. Other moving objects outside that ROI may still have valid velocities.

Geometry budget stress

Budget	Captured vertices	Capped meshes
262K	262,144	645
1M (new default)	370,840	0
2M	370,840	0

The isolated scene contained 2,200 extra pigs. Partial meshes are rejected instead of matching a truncated vertex sequence. This is a deliberate extreme-density test, not an FPS benchmark. The counters describe intercepted submissions; they cannot prove coverage of every renderer.

One velocity per pixel remains a representation limit: overlapping independently moving translucent layers cannot all be exact simultaneously. The opacity control selects the dominant contributor; newly born particles have no previous geometry. The patch preserves normal world/OIT ordering rather than drawing particles through water or walls.

MEASURED CORRECTNESS

Camera, color and lifecycle

Camera motion test	Samples	95th-percentile error
Yaw	88462	0.001002 input pixels
Pitch	89940	0.000467 input pixels
Strafe	88120	0.000753 input pixels

CPU ground truth independently reconstructs positions from the captured D32 depth and jittered inverse projection, then compares previous/current unjittered projection with RG16F vectors. These subpixel errors are consistent with half-float storage; the established native Y-jitter boundary is preserved.

Native color probes

Encoding / source patch	Measured output	Largest channel difference
SDR: four dark/saturated patches	Matches source quantized UNORM values	0 at sampled centers
Linear: (0.001, 0.001, 0.001)	(0.0009985, 0.0009985, 0.0009985)	0.000000954
Linear: (0.18, 0.5, 0.9)	(0.17969, 0.5, 0.89990)	0.000244
Linear HDR: (4, 2, 0.5)	(3.99805, 1.99902, 0.50049)	0.001953

Source and output were read directly from GPU buffers after real DLSS dispatches. Four settled captures were taken per patch. This verifies precision/range and the matching processing flag, not arbitrary shader-pack exposure or HDR display correctness. Forcing linear processing on vanilla gamma-encoded color is not a valid visual-quality comparison.

Passed scenarios

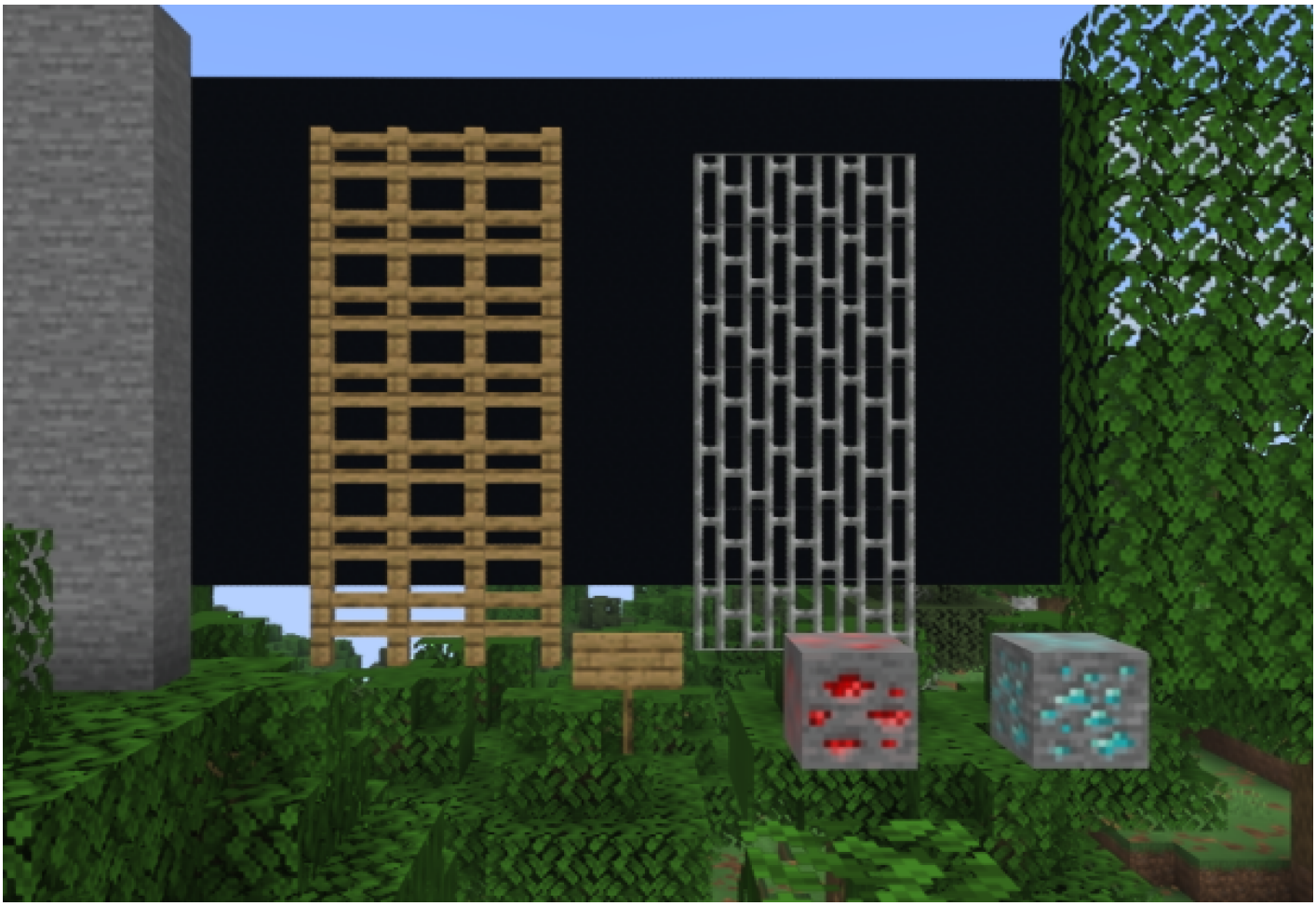
K/M/L/Auto requests; Quality, Performance, Ultra Performance and native DLAA; 720p, 1080p and 1440p output; yaw/pitch/strafe; moving pig and chest; opaque and translucent particles; OIT on/off; depth occlusion; first/third-person views; FOV changes; resource reload; overworld/nether transition; render-gap reset; native SDR/HDR feature switches; injected failure/retry; FSR2/FSR3/TAA shared-path regression checks.

Khronos core validation plus synchronization validation were enabled for the final release process using VK_VALIDATION_VALIDATE_SYNC=1. Final validation results and release hashes are recorded in the evidence JSON. Earlier failed development launches are excluded from the successful capture set. NVIDIA logs retain manual-hook/private-data warnings associated with the manual Vulkan 1.2 integration; no claim is made that those informational SDK warnings vanish.

Automated verification: 142 tests, 0 failures, 0 errors. Public JAR: own DLSS bridge, own FSR bridge and AMD runtime only. No NVIDIA DLLs or disposable test harness. Normal runClient config/options are byte-identical to their backups.

Final process result: zero Vulkan VUID errors, zero synchronization hazards and no leaked objects at device destruction. World unload separately reported zero owned AA texture bytes and zero hand-AA texture bytes. An earlier exit check exposed five cached pass buffers; their cleanup was added and the full device exit retested.

Texture detail without blanket bias



Quality, MIP bias Off. Enlarged crop includes thin fence/bar edges, alpha-cutout leaves, ore textures and sign geometry. Native DLAA and moving Quality Off/Automatic sequences were captured for comparison.

A 64px high-frequency checker resource pack was loaded and removed in the sandbox. Its pattern remained visible; pack reload and return to the vanilla atlas worked. Texture bias remains optional and terrain-specific. Negative bias can recover detail but also expose undersampling and shimmer; this test does not justify forcing it on every texture family or replacing the existing default.

Compatibility limits

Standard vanilla perceptual SDR is the default supported contract. Auto assumes floating-point upstream targets are linear; renderers that use a different encoding must select the correct override. External pre-exposure, custom tone mapping and shader-pack attachments still require a matching renderer adapter. The control does not implement HDR display output.

Captured geometry requires stable identity and vertex order. New topology, births, special text/effects and renderers bypassing the intercepted feature paths may retain camera flow. The expanded counters expose intercepted gaps; no claim is made of universal renderer coverage. DLSS model requests can be overridden by NVIDIA App. Validation on this GPU cannot certify every driver/GPU combination or guarantee artifact-free output in every scene.

Release and evidence

Release artifact

build/libs/salts-anti-aliasing-mc26.3-1.1.jar

SHA-256: 67f3bde273b7cd5b244c21798671df278d4e55023b0f72dcbb438ab631347377

The report is paired with compact evidence files: numeric camera/particle/color measurements, source/request metadata, build and validator logs, release packaging checks, and representative images. Full raw buffers and control scripts remain under build/dlss-integration-fixes-20261025 and run/dlss-integration-fixes-20261025 in this workspace.

Validation controls

The test harness is a separate development mod. All native-dispatch captures assert a healthy rendering status and use unique labels. Screenshots are taken after warmup. Camera error uses native buffer coordinates; visual heatmaps flip rows once to match Minecraft screenshots. Temporal comparisons use zero extra sharpening. The HDR patch tests inspect floating-point output before the final display conversion.

Primary integration references

NVIDIA DLSS programming guide

Streamline DLSS integration and required inputs

Pinned Streamline 2.14.1 release

Khronos synchronization validation setup

Local sources: corrected 25 October integration audit; patched Java/JNI/shaders; captured GPU buffers, isolated game screenshots, native logs and final Gradle results. Recommendations and compatibility boundaries follow the actual source encoding, supported model contracts and observed data.